Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Consider the following statements
(1) Mode can be computed from histogram
(2) Median is not independent of change of scale
(3) Variance is independent of change of origin and scale.
Which of these is/are correct?

(A) (1), (2) and (3)

(B) Only (2)

(C) Only (1) and (2)

(D) Only (1)

Q.2

If in a frequency distribution, the mean and median are 21 and 22 respectively, then the mode is approximately

(A) 25.5

(B) 24.0

(C) 22.0

(D) 20.5

Q.3

The mean of the data set comprising of 16 observations is 16. If one of the observation valued 16 is deleted and three new observations valued 3,4 and 5 are added to the data, then the mean of the resultant data, is :

(A) 15.8

(B) 14.0

(C) 16.8

(D) 16.0

Q.4

The mean of the numbers $a, b, 8, 5, 10$ is 6 and the variance is 6.80. Then which one of the following gives possible values of a and b ?

(A) $a = 0, b = 7$

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(B) $a=5, b=2$

(C) $a=1, b=6$

(D) $a=3, b=4$

Q.5

The mean of n items is $\bar{x}$. If the first term is increased by 1 , second by 2 and so on, then new mean is

(A) $\bar{x}+n$

(B) $\bar{x}+\frac{n}{2}$

(C) $\bar{x}+\frac{n+1}{2}$

(D) None of these

Q.6

The mean weight per student in a group of seven students is $55 \mathrm{~kg}$. If the individual weights of six students are $52, 58, 55, 53, 56$ and 54 , then the weight of the seventh student is

(A) $55 \mathrm{~kg}$

(B) $60 \mathrm{~kg}$

(C) $57 \mathrm{~kg}$

(D) $50 \mathrm{~kg}$

Q.7

The scores of 15 students in an examination were recorded as $10, 5, 8, 16, 18, 20, 8, 10, 16, 20, 18, 11$, 16,14 and 12. After calculating the mean, median and mode, an error is found. One of the values is wrongly written as 16 instead of 18 . Which of the following measures of central tendency will change?

(A) Mean and median

(B) Median and mode

(C) Mode only

(D) Mean and mode

Q.8

For the data $3, 5, 1, 6, 5, 9, 5, 2, 8, 6$ the mean, median and mode are x, y and z respectively. Which one of the following is correct?

(A) $x=y \neq z$

(B) $x \neq y=z$

(C) $x \neq y \neq z$

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(D) $x = y = z$

Q.9

If mean of the n observations $x_1, x_2, x_3, \ldots x_n$ be $\bar{x}$ then the mean of n observations $2x_1 + 3, 2x_2 + 3, 2x_3 + 3, \ldots., 2x_n + 3$ is

(A) $3\bar{x} + 2$

(B) $2\bar{x} + 3$

(C) $\bar{x} + 3$

(D) $2\bar{x}$

Q.10

A school has four sections of chemistry in class XII having $40, 35, 45$ and 42 students. The mean marks obtained in Chemistry test are $50, 60, 55$ and 45 respectively for the four sections, the over all average of marks per students is

(A) 53

(B) 45

(C) 55.3

(D) 52.25

Q.11

The mean of five numbers is 30 . If one number is excluded, their mean becomes 28 . The excluded number is

(A) 28

(B) 30

(C) 35

(D) 38

Q.12

If the arithmetic mean of the numbers
$x_1, x_2, x_3, \ldots \ldots x_n$ is $\bar{x}$. then the arithmetic mean
ofnumbers $ax_1 + b, ax_2 + b, ax_3 + b, \ldots . .ax_n + b$, where a, b are two constants would be

(A) $\bar{x}$

(B) $na\bar{x} + nb$

(C) $a\bar{x}$

(D) $a\bar{x} + b$

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Q.13

The average of n numbers $x_1, x_2, x_3, \ldots\ldots, x_n$ is M If x_n is replaced by x', then new average is

(A) $\mathrm{M} - x_n + x'$

(B) $\frac{nM - x_n + x'}{n}$

(C) $\frac{(n-1)M + x'}{n}$

(D) $\frac{M - x_n + x'}{n}$

Q.14

If the mean of the numbers $27 + x, 31 + x, 89 + x, 107 + x, 156 + x$ is 82 , then the mean of $130 + x, 126 + x, 68 + x, 50 + x, 1 + x$ is

(A) 75

(B) 157

(C) 82

(D) 80

Q.15

Mean of the numbers $1, 2, 3, \ldots, n$ with respective weights $1^2 + 1, 2^2 + 2, 3^2 + 3, \ldots, n^2 + n$ is

(A) $\frac{3n(n+1)}{2(2n+1)}$

(B) $\frac{2n+1}{3}$

(C) $\frac{3n+1}{4}$

(D) $\frac{3n+1}{2}$

Q.16

The mean income of a group of 50 persons was calculated as ₹169. Later it was discovered that one figure was wrongly taken as 134 instead of correct value 143 . The correct mean should be (in ₹)

(A) 168

(B) 169

(C) 168.92

(D) 169.18

Q.17

The variance of first 100 odd natural numbers is

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(A) 2222

(B) 3333

(C) 4444

(D) 5555

Q.18

Consider any set of observations $x_1, x_2, x_3, \ldots, x_{101}$; it being given that $x_1 < x_2 < x_3 < \ldots < x_{100} < x_{101}$; then the mean deviation of this set of observations about a point k is minimum when k equals

(A) x_1

(B) x_{51}

(C) $\frac{x_1+x_2+\ldots+x_{101}}{101}$

(D) x_{50}

Q.19

Given (i) 85 observations which are not sorted and (ii) 150 observations which are sorted and arranged in an increasing order. The median values of (i) & (ii) respectively can be found as

(A) (i) 43rd observation (ii) A.M. of 75th and 76th observation

(B) (i) 43rd observation (ii) 76th observation

(C) (i) can not be found (ii) can not be found

(D) None of these

Q.20

In a binomial distribution, the mean is 4 and the variance is 3. What is the mode?

(A) 6

(B) 5

(C) 4

(D) 3

Answers & Solutions

Q.1 Answer:

Only (1) and (2)

Solution:

Mode is the term which has the maximum frequency. Hence, it can be calculate using a histogram.

Median is the medium value of the terms. Hence, it is independent of the change of scale.

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Variance depends on the data, if the scale changes, the data will change and variance of the data will change.

Q.2 Answer:

24.0

Solution:

Mode $= 3$ **Median** -2 **mean**

$= 3 \times 22 - 2 \times 21$

$= 24$

Q.3 Answer:

14.0

Solution:

$\Sigma x_i = 256$

Let new observations be X_i

$\Sigma X_i = (256 - 16) + (3 + 4 + 5) = 252$ & **number of observations are 18**

$\therefore$ **New mean** $= \frac{\Sigma X_i}{18} = 14$

Q.4 Answer:

$a = 3, b = 4$

Solution:

Mean of $a, b, 8, 5, 10$ **is 6** $\Rightarrow \quad \frac{a+b+8+5+10}{5} = 6 \Rightarrow a + b = 7$

Variance of $a, b, 8, 5, 10$ **is** 6.80

$\Rightarrow \frac{(a-6)^2+(b-6)^2+(8-6)^2+(5-6)^2+(10-6)^2}{5}$

$= 6.80$

$\Rightarrow \quad a^2 - 12a + 36 + (1 - a)^2 + 21 = 34$

[using eq. $\Rightarrow \quad 2a^2 - 14a + 24 = 0 \Rightarrow a^2 - 7a + 12 = 0$

$\Rightarrow a = 3$ **or** $4 \Rightarrow b = 4$ **or 3**

The possible values of a **and** b **are** $a = 3$ **and** $b = 4$ **or,** $a = 4$ **and** $b = 3$

Q.5 Answer:

$\bar{x} + \frac{n+1}{2}$

Solution:

Let $x_1, x_2 \ldots x_n$ **be** n **items. Then,** $\bar{x} = \frac{1}{n}\Sigma x_i$ **Let** $y_1 = x_1 + 1, y_2 = x_2 + 2, y_3 = x_3 + 3, \ldots, y_n = x_n + n$

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Then the mean of the new series is

$$\frac{1}{n}\Sigma y_i = \frac{1}{n}\sum_{i=1}^{n}\left(x_i + \frac{1}{i}\right)$$
$$= \frac{1}{n}\sum_{i=1}^{n} x_i + \frac{1}{n}(1+2+3+\ldots..+n)$$
$$= \bar{x} + \frac{1}{n}\cdot\frac{n(n+1)}{2} = \bar{x} + \frac{n+1}{2}.$$

Q.6 Answer:

57 kg

Solution:

The total weight of seven students is $55 \times 7 = 385$ kg

The sum of the weights of six students is $52 + 58 + 55 + 53 + 56 + 54 = 328$ kg

Hence, the weight of the seventh student is $385 - 328 = 57$ kg

Q.7 Answer:

Mean and mode

Solution:

Mean of the scores $= \frac{202}{15}$

Mean of the correct scores $= \frac{200}{15}$

i.e., Mean changes.

Median is same for both cases i.e., 14

Mode is proportional to mean.

Q.8 Answer:

$x = y = z$

Solution:

Given data $3, 5, 1, 6, 5, 9, 5, 2, 8, 6$ **and mean, median and mode are** x, y, z **respectively.**

Rearranging data $1, 2, 3, 5, 5, 5, 6, 6, 8, 9$

Mean $= x = \frac{1+2+3+5+5+5+6+6+8+9}{10} = \frac{50}{10} = 5$

Median $= y = \frac{\frac{n^{\text{th}}}{2}\text{ term} + \left(\frac{n}{2}+1\right)^{\text{th}}\text{ term}}{2}$

$y = \frac{5+5}{2} = 5$

Mode (z) $=$ **most frequently occuring value** $= 5$

Hence $x = y = z$.

Q.9 Answer:

$2\bar{x} + 3$

Solution:

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Required mean $= \frac{1}{n}\sum_{i=1}^{n}(2x_i+3)$
$= \frac{2}{n}\left(\sum_{i=1}^{n}x_i\right)+\frac{3n}{n} = 2\left\{\frac{1}{n}\left(\sum_{i=1}^{n}x_i\right)\right\}+3$
$= 2\bar{x}+3$

Q.10 Answer:

52.25

Solution:

Total number of students $= 40+35+45+42 = 162$
Total marks obtained $= (40\times 50)+(35\times 60)+(45\times 55)+(42\times 45) = 8465$
Overall average of marks per students
$= \frac{8465}{162} = 52.25$

Q.11 Answer:

38

Solution:

Mean of 5 numbers $= 30$
$\therefore$ **Total sum of 5 numbers** $= 30\times 5 = 150$ **After excluded one number Mean of 4 numbers will be** $= 28$
$\therefore$ **Total sum of 4 numbers** $= 4\times 28 = 112$ **Thus, excluded number**
$=$ (**sum of 5 numbers** $-$ **sum of 4 numbers)** $= 150-112 = 38$

Q.12 Answer:

$a\bar{x}+b$

Solution:

Requiredmean $= \frac{(ax_1+b)+(ax_2+b)+\ldots..+(ax_n+b)}{n}$
$= \frac{a(x_1+x_2+\ldots..+x_n)+nb}{n} = a\bar{x}+b$
$\left(\because \frac{x_1+x_2+\ldots..+x_n}{n} = \bar{x}\right)$

Q.13 Answer:

$\frac{nM-x_n+x'}{n}$

Solution:

$M = \frac{x_1+x_2+x_3\ldots\ldots x_n}{n}$
$nM = x_1+x_2+x_3+\ldots\ldots+x_{n-1}+x_n$
i.e., $nM-x_n = x_1+x_2+x_3+\ldots\ldots+x_{n-1}$
$\frac{nM-x_n+x'}{n} = \frac{x_1+x_2+x_3+\ldots\ldots x_{n-1}+x'}{n}$
$\therefore$ **New average** $= \frac{nM-x_n+x'}{n}$

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Q.14 Answer:

75

Solution:

Given $82 = \frac{(27+x)+(31+x)+(89+x)+(107+x)+(156+x)}{5}$

$\Rightarrow \quad 82 \times 5 = 410 + 5x \Rightarrow 410 - 410 = 5x \Rightarrow x = 0$

Required mean is, $\bar{x} = \frac{130+x+126+x+68+x+50+x+1+x}{5}$

$\bar{x} = \frac{375+5x}{5} = \frac{375+0}{5} = \frac{375}{5} = 75$

Q.15 Answer:

$\frac{3n+1}{4}$

Solution:

Here for each $x_i = 1$**, weight** $w_i = i^2 + i$

Hence, the required mean

$= \frac{\sum w_i x_i}{\sum w_i} = \frac{\sum_{i=1}^{n} i\left(i^2+i\right)}{\sum_{i=1}^{n}(i^2+i)}$

$= \frac{\sum_{i=1}^{n} i^3+\sum_{i=1}^{n} i^2}{\sum_{i=1}^{n} i^2+\sum_{i=1}^{n} i}$

$= \frac{\frac{n^2(n+1)^2}{4}+\frac{n(n+1)(2n+1)}{6}}{\frac{n(n+1)(2n+1)}{6}+\frac{n(n+1)}{2}}$

$= \frac{\frac{n(n+1)}{2}\left\{\frac{n(n+1)}{2}+\frac{2n+1}{3}\right\}}{\frac{n(n+1)}{2}\left\{\frac{2n+1}{3}+1\right\}}$

$= \frac{3n^2+7n+2}{2(2n+4)} = \frac{(3n+1)(n+2)}{4(n+2)} = \frac{3n+1}{4}$

Q.16 Answer:

169.18

Solution:

Given, $169 = \frac{\Sigma x}{50}$

Correct value is 143**. Hence, sum is short by 9 .**

$\therefore M = \frac{169 \times 50+9}{50} = 169 + \frac{9}{50} = 169.18$

Q.17 Answer:

3333

Solution:

Variance $= \frac{1^2 + 3^2 + 5^2 + \ldots + \left(199\right)^2}{100} - \frac{1 + 3 + 5 + \ldots + 199^2}{100}$

Sum $= \sum_{n=1}^{n} \left(2n - 1\right)^2 = 4\Sigma n^2 - 4\Sigma n + n$

Sum $= \frac{4n(n+1)(2n+1)}{6} - \frac{4n(n+1)}{2} + n$

Put $n = 100$

$= \frac{4 \times 100 \times 101 \times 201}{6} - 2 \times 100 \times 101 + 100$

$= 100\frac{2 \times 101 \times 201}{3} - 2 \times 101 + 1 = 100\left(2 \times 6767 - 202 + 1\right)$

$= 100\left(13534 - 201\right) = 100 \times 13333 = 1333300$

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Then, Variance $= \frac{1333300}{100} - \left(100\right)^2 = 13333 - 10000 = 3333$

Q.18 Answer:

x_{51}

Solution:

Mean deviation is minimum when it is considered about the item, equidistant from the beginning and the end i.e. the median. In this case median is $\frac{101+1}{2}$ **th i.e. 51 st item i.e.,** x_{51}**.**

Q.19 Answer:

None of these

Solution:

For the second set of observations $N = 150$ **(even), so median is the A.M. of** 75th **and** 76th **observation first set is not sorted so we have to arrange them in increasing or decreasing order**

Q.20 Answer:

4

Solution:

As give, $np = 4$ **and** $npq = 3$ **[where p is the probability of success and q is the probability of failure for an event to occur, and 'n' is the number of trials]**

$\Rightarrow \quad q = \frac{npq}{np} = \frac{3}{4}$

Also, $p = 1 - q = 1 - \frac{3}{4} = \frac{1}{4}$

$\therefore \quad n = 16$

In a binomial distribution, the value of r **for which** $P(X = r)$ **is maximum is the mode of binomial distribution.**

hence, $(n+1)p - 1 \leq r \leq (n+1)p$

$\Rightarrow \quad \frac{17}{4} - 1 \leq r \leq \frac{17}{4} \Rightarrow \frac{13}{4} \leq r \leq \frac{17}{4}$

$\Rightarrow \quad 3.25 \leq r \leq 4.25$

$\Rightarrow \quad r = 4$

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